Contribution Title

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Keywords: First keyword · Second keyword · Another keyword.

1 Introduction

Prostate-specific membrane antigen (PSMA) PET/CT has become a central imaging modality for staging, treatment planning, radiopharmaceutical therapy, and post-therapy response assessment in prostate cancer [**?**]. In these settings, patients often undergo multiple scans over time, comprising a baseline scan before therapy and one or more follow-up scans afterwards; accurate baseline-to-follow-up registration can support comparison of disease burden, localisation of uptake changes, assessment of lesion progression or response, dose accumulation, and quantitative longitudinal analysis [**?**]. The challenge organisers identify

several factors that make establishing this correspondence by deformable image registration difficult in routine whole-body data [**?**]: baseline and follow-up scans differ substantially in field of view, patient positioning, respiratory state, bladder and bowel filling, and body weight; CT and PET encode complementary but appearance-dissimilar information; and the registration must preserve the quantitative PET signal used for response assessment while remaining physically plausible structures such as the skeleton. The Learn2Reg 2026 PSMAReg challenge [**?**,**?**,**?**] formalises this problem as the estimation of a dense displacement field that warps each follow-up (moving) scan to its baseline (fixed) scan, evaluated jointly on organ overlap, boundary distance, deformation regularity, and preservation of PET-derived tumour biomarkers.

Deformable registration has traditionally been posed as an iterative optimisation over a regularised similarity objective, with widely used instances including the diffeomorphic Demons [**?**], SyN [**?**], and B-spline [**?**] schemes. While accurate, these methods are slow at whole-body resolution. Learning-based methods instead amortise the optimisation over a training set into the network weights, replacing per-pair iterative optimisation with a single forward pass at inference. VoxelMorph established the unsupervised paradigm [**?**], and subsequent work introduced diffeomorphic parameterisations through stationary velocity fields [**?**], transformer backbones such as TransMorph [**?**], and multi-resolution architectures such as LapIRN [**?**]. Among the latter, LapIRN uses a Laplacian pyramid of coarse-to-fine sub-networks to recover large deformations while keeping the transformation smooth and invertible, and has been a strong performer across previous Learn2Reg editions [**?**,**?**]. A recurring limitation of purely intensity-driven losses is that they permit non-physical folding. Anatomy-aware regularisation that penalises non-rigid deformation, e.g. of skeletal structures [**?**,**?**,**?**,**?**], is therefore particularly relevant for whole-body PET/CT, where the skeleton is a frequent site of PSMA-avid disease.

In this work, we adapt LapIRN to longitudinal whole-body PSMA PET/CT for the PSMAReg task. We retain its diffeomorphic, coarse-to-fine formulation to handle the large field-of-view shifts between visits, and augment it with several task-specific regularisation terms targeted at the challenge objectives: a bone-rigidity loss that discourages implausible deformation of the skeleton, a total-lesion-glycolysis (TLG) loss that preserves PET-derived lesion burden under the estimated deformation, and a term based on the Jacobian determinant within the tumour volume that acts as a proxy for preserving the metabolic tumour volume (MTV). **[Additional contributions to be added.]** We describe the method and the training and inference infrastructure, and report validation leaderboard results for each metric.

2 Methods

2.1 Data

2.2 Preprocessing

2.3 Registration Pipeline

2.4 Losses

DSC Loss We used a standard DSC loss for weak label supervision. Since the challenge scores PET organ overlap but provides no PET organ annotations for training, we measured each of the 117 TotalSegmentator organs' mean SUV relative to skeletal muscle across all sessions. As 90 organs fall below twice the muscle background and are thus indistinguishable in PET, we upweighted only the 10 exceeding 4 times the background (kidneys, gallbladder, spleen, liver, duodenum, bladder, portal vein, prostate, pancreas) by a factor of 2 in the per-class Dice loss. This relies on PET and CT being well aligned within a session, so that CT-derived overlap on these organs approximates the PET organ overlap being scored.

Preservation of PET quantification The scored MTV bias only constrains the *net* lesion volume, so we penalize it directly while supplying dense gradients through a Jacobian-based surrogate. All terms are evaluated on the total transformation $\varphi = \varphi_{\text{aff}} \circ \varphi_\theta$, i.e. the affine pre-registration composed with the predicted displacement, against the original moving lesion mask M, since the affine itself is generally not volume-preserving. First, we minimize the squared relative volume error

$$\mathcal{L}_{\text{MTV}} = \left(\frac{|M \circ \varphi| - |M|}{|M|} \right)^2, \tag{1}$$

where $M \circ \varphi$ denotes the linearly resampled mask. This matches the scored metric exactly, but its gradient is confined to the lesion's sub-voxel boundary shell. We therefore add a complementary term based on the change-of-variables identity: summing $\det J_\varphi$ over the warped lesion region recovers the original lesion volume, so the mean determinant over the warped lesion equals the ratio of original to warped volume. Driving this mean to one,

$$\mathcal{L}_{\text{MTV-J}} = \left(\frac{1}{|W|} \sum_{x \in W} \det J_\varphi(x) - 1 \right)^2, \tag{2}$$

where W denotes the warped lesion region, therefore pins the same net volume ratio as (**??**) while providing gradients throughout the lesion interior; unlike a per-voxel constraint, it still permits internal redistribution of volume. Note that masking the determinant with the moving-frame lesion, as is tempting, samples $\det J_\varphi$ at locations unrelated to the lesion once the affine is composed

in; empirically, this decorrelated the surrogate from the scored MTV. Finally, the per-voxel penalty

$$\mathcal{L}_{\text{jac}} = \frac{1}{|W|} \sum_{x \in W} \left(\det J_\varphi(x) - 1\right)^2 \tag{3}$$

acts as a stricter local regulariser, discouraging compensating compression and expansion within the lesion.

The challenge additionally scores the total lesion glycolysis (TLG), the PET-activity–weighted counterpart of the MTV, which we penalize analogously as the relative change of intensity mass under the warp:

$$\mathcal{L}_{\text{TLG}} = \frac{\left|\sum_x (I \circ \varphi)(x)\,(M \circ \varphi)(x) - \sum_y I(y)\,M(y)\right|}{\sum_y I(y)\,M(y)}, \tag{4}$$

where I denotes the moving PET image. Since both the image and the mask are resampled with the same transformation, this term is sensitive not only to volume change but also to intensity blurring at the lesion boundary, and thus complements the purely geometric terms (**??**)–(**??**).

The same terms (**??**)–(**??**) are used during training and descended at test time by our instance optimisation, so the network is trained on the objective it is later refined under.

Displacement field regularisation To regularise the displacement field, we used a diffusion regulariser and a Jacobian loss that penalises non-diffeomorphic voxels.

In 146 patients with paired baseline and follow-up scans, a median of 46.0% (IQR 3.1 to 72.3%) of total lesion volume at baseline lay within skeletal structures, rising to 63.8% (IQR 12.1 to 77.4%) at follow-up. Since a substantial fraction of lesion volume is therefore intraosseous, we added the local rigidity penalty of Staring et al.[staring2007] [TODO mention all papers and the 3 losses we used for rigidity] on a bone mask as an anatomically grounded surrogate: enforcing rigid bone motion is both physiologically correct and, where lesions sit in bone, directly prevents the volume change MTV measures. To test whether accuracy and preservation genuinely conflict, we logged the pairwise cosine similarities of all per-term gradients with respect to the shared parameters, along with their norms and shares. The groups were persistently opposed (cos = -0.25, negative on 80% of steps) at comparable magnitude (ratio of norms about 1.0), excluding weight starvation. Attribution localised nearly all of it to one term: rigidity contributed -0.20 of the -0.25, whereas TLG and the lesion Jacobian contributed about 0.00 despite holding 45% and 5% of their group's norm. TLG was orthogonal to every term (|cos| below 0.05), that is, loud, unopposed and still underperforming, indicating a proxy problem rather than a conflict. Rigidity instead behaved as a regulariser (cos = +0.53 with smoothness, -0.53 with NCC, 0.00 with TLG), so the apparent accuracy-versus-preservation trade-off was, in fact, a regulariser-versus-data-term tension.

Rigidity opposed bone Dice (-0.45) more than soft tissue Dice (-0.23), which no correct bone penalty should do. Its three conditions are evaluated by finite differences at every voxel and only then masked, so the mask selects which voxels are scored but not which are read. The stencil used for the derivative calculation would therefore also use non-bone voxels. With thin cortical bone and ribs, this dominates. We tested it across five cases: 53.0% (range 51.6 to 54.7) of the voxels entering the loss were not bone; a field with zero displacement inside every bone scored 56 (range 40 to 69), with 100% removed by a single mask erosion; and 44.8% of the gradient magnitude fell outside bone. The term measured the bone-to-soft tissue displacement discontinuity, which any correct registration must exhibit. Erosion of the mask could therefore fix this problem; however, in the current dataset, many bone masks are only a few voxels wide. We therefore replaced this loss with a per-structure rigid-fit residual: for each label, we obtain the optimal rotation and translation from the predicted displacement in closed form via Kabsch alignment, and penalise the mean squared residual of that fit. No stencil crosses the mask, and each label is fitted independently, so articulation is free. The zero displacement field now scores exactly 0 and a per-bone rigid field 2.4e-6, while anisotropic squeeze is still rejected at 1.16. The penalty constrains only the shape of the deformation within each structure and is invariant to its placement, leaving alignment to NCC and Dice, unlike derivative based penalties [staring2007, modersitzki2008] and unlike Förner and Wendler [foerner2025], who regress the field onto a per bone rigid field precomputed by Dice pre registration, requiring segmentations in both images and inheriting pre registration error as a training target.

No surface distance loss We deliberately do not add a differentiable surface-distance term (e.g. the distance-transform Hausdorff surrogate of Karimi and Salcudean [TODO CITATION]) to either the training objective or the instance optimisation. A per-label analysis of the validation predictions shows that the median and the 75th percentile of the per-label HD95 both equal 2.73 mm, i.e. exactly the in-plane voxel spacing: at least three quarters of all labels already lie at the discretisation floor attainable by a nearest-neighbour–warped label map. The mean HD95 is instead driven almost entirely by a small tail, with the worst 10% of (case, label) pairs contributing 57.6% of the total. This tail is dominated by structures for which no boundary refinement is meaningful: partially visible field-of-view–edge anatomy (brain, skull), limb structures subject to inter-session repositioning (humeri, scapula), and hollow organs with variable filling and hence no well-defined point correspondence (bowel, colon, urinary bladder). A boundary loss would therefore expend its gradient on labels that are already optimal, while leaving the tail — a capture-range and correspondence problem rather than a boundary-precision one — unaffected.

2.5 Model Selection

Model selection enters our pipeline twice: once *within* a run, to decide which checkpoint to keep, and once *across* runs, to decide which variant to submit

in the container. Both use the same three-component quality notion, and differ only in how each component is scored.

Both stages combine registration accuracy (CT DSC and HD95), PET-biomarker preservation (absolute MTV and TLG bias) and deformation regularity (%NDV after the official 0.005% practical-equivalence adjustment) as a weighted geometric mean with exponents 0.4, 0.4 and 0.2, mirroring the challenge's final score

$$S = 100 \cdot A^{0.4}\, B^{0.4}\, R^{0.2}, \tag{5}$$

where the accuracy term A is the geometric mean of the DSC and HD95 scores, the biomarker term B that of the MTV and TLG scores, and R the %NDV score.

Checkpoint selection. Registration accuracy and tumour preservation do not peak together: in level-3 training, validation Dice improves until it plateaus, whereas the tumour metrics reach their optimum roughly halfway through and degrade thereafter. Selecting on Dice alone therefore yields a checkpoint well past the tumour optimum, so we select on Eq. (**??**) instead. The official per-metric scores are ranks within a cohort of competing methods, which does not exist during a single training run. We therefore replace each rank score with a logistic quality centred on the metric's median and scaled by its robust spread (interquartile range divided by 1.349), both measured over the validation rounds of a completed run.

Submission selection. Choosing between trained variants does provide a cohort, so here we evaluate Eq. (**??**) exactly as the challenge does: each of the five metrics is ranked separately across the K variants, with ties taking the worst rank they span, and rank r is turned into the score $(K - r + 1)/K$. The variant with the highest S is the one we submit. To check that this choice is not an artefact of a few cases, we additionally compare the selected variant against every other variant per metric with a paired Wilcoxon signed-rank test over the validation cases, Holm-corrected within each metric. As the variant is selected on the same cases, these tests are exploratory rather than confirmatory; the official validation leaderboard serves as the held-out check.

2.6 Baselines

3 Results

3.1 Evaluation

The challenge does not release labels for the validation cases. Unless stated otherwise, the validation numbers we report are therefore computed against surrogate labels we generated ourselves: CT organ labels with TotalSegmentator [**?**] and

PET tumour labels with the winning autoPET-III submission [**?**]. These surrogate labels underlie every validation metric in this work, including the checkpoint and submission selection of Sec. **??**. Being predictions rather than the challenge's reference annotations, they carry their own error, so their absolute values are not directly comparable to the official validation leaderboard; we use them to compare methods against each other under an identical protocol. Where we report official validation-leaderboard results, these are computed by the organisers against their reference annotations and are marked as such. Since the leaderboard covers the same cases with the correct labels, we use the agreement between the two rankings as a check on the surrogate protocol, and prefer the leaderboard whenever it is available for the models being compared.

Table **??** compares the nine LapIRN variants. V5 attains the highest challenge score (69.0), with 88.3% DSC, 4.77 mm HD95 and MTV and TLG errors of 1.83% and 1.80%. Its registration accuracy is significantly better than that of every other variant except V3, which reaches a marginally higher DSC (88.9%) but degrades the PET biomarkers by an order of magnitude (15.0% MTV and 16.2% TLG error), illustrating that accuracy and biomarker preservation trade off against each other. Six variants produce fewer non-diffeomorphic voxels than V5, but all of them lie below the 50 ppm practical-equivalence threshold, so the difference does not affect the ranking.

Table 1. LapIRN variants on the official challenge validation set ($n = 20$ cases), evaluated against the surrogate labels of Sec. **??**. Each cell reports the mean over cases. DSC and the MTV/TLG errors are given in %, NDV in ppm. Score is the challenge score of Eq. (**??**); variants are listed in ranking order, highest score first. Bold marks the best value in each column, and the Model and Score entries of the best-scoring variant. * marks metrics on which that variant is significantly better than the variant in the given row (paired Wilcoxon signed-rank, Holm-corrected within each metric, $p < 0.05$).

Model	DSC (%) ↑	HD95 (mm) ↓	MTV %err ↓	TLG %err ↓	NDV (ppm) ↓	Score ↑
V5	88.3	4.77	1.83	1.80	20.7	**69.0**
V6	87.6*	4.93*	1.94	1.64	19.3	63.4
V4	87.5*	5.02*	1.89	**1.64**	6.49	60.3
V9	88.2	4.99	**1.11**	2.14	96.3*	57.7
V2	83.4*	6.08*	2.78	1.79	0.375	43.7
V3	**88.9**	**4.69**	15.0*	16.2*	7.72	33.3
V1	87.2*	5.15*	7.98*	4.99*	156*	29.3
V8	78.8*	6.54*	7.96*	12.2*	7.64	26.1
V7	75.0*	7.00*	2.93	5.98*	**0.00**	22.8

Table **??** places the selected variant against the baselines. It improves DSC from 73.3% (NiftyReg, the strongest baseline) to 88.3% and halves HD95 from 7.09 to 4.77 mm, while reducing the MTV and TLG errors by more than a factor of two and the non-diffeomorphic volume from 1525 to 21 ppm. All differences are significant on every metric (paired Wilcoxon signed-rank, Holm-corrected, $p < 0.01$).

Table 2. The selected variant against the baselines on the official challenge validation set ($n = 20$ cases), evaluated against the surrogate labels of Sec. **??**. Each cell reports the mean over cases. DSC and the MTV/TLG errors are given in %, NDV in ppm. Bold marks the best value in each column. * marks metrics on which our method is significantly better than the baseline in the given row (paired Wilcoxon signed-rank, Holm-corrected within each metric, $p < 0.05$). Dashes mark metrics that are undefined for a baseline: the unregistered and affine cases have no deformation field, and the PET biomarkers are only meaningful after resampling.

Method	DSC (%) ↑	HD95 (mm) ↓	MTV %err ↓	TLG %err ↓	NDV (ppm) ↓
Initial	14.8*	30.9*	–	–	–
Affine	52.9*	11.7*	5.47*	4.70*	–
NiftyReg	73.3*	7.09*	4.46*	8.68*	1525*
ConvexAdam	63.0*	8.41*	5.10*	4.15*	100*
Ours	**88.3**	**4.77**	**1.83**	**1.80**	**20.7**

4 Discussion

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Disclosure of Interests. It is now necessary to declare any competing interests or to specifically state that the authors have no competing interests. Please place the statement with a bold run-in heading in small font size beneath the (optional) acknowledgments[4], for example: The authors have no competing interests to declare that are relevant to the content of this article. Or: Author A has received research grants from Company W. Author B has received a speaker honorarium from Company X and owns stock in Company Y. Author C is a member of committee Z.

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